Feature-based Image Matching and Automatic Panorama Construction

Submission packet

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Abstract

This packet summarises a classical OpenCV panorama pipeline (detect → describe → match → RANSAC → homography → warp → stitch), with source excerpts, experiment figures, and screenshots of the interactive app. The full CVPR-formatted report, code, and data live in the public repository. The live demonstration is hosted on Streamlit Community Cloud.

1. Where to find the work

- Live app: https://feature-based-panorama.streamlit.app/
- Source (public): https://github.com/qbentil/feature-based-panorama
- Full report PDF in the repo: report/feature_based_panorama.pdf (also embedded on the app's Project report tab)

The repository contains the pipeline under src/, the Streamlit walkthrough under app_pages/, Wikimedia Commons campus photographs (Balme Library, Dance Department, Night Market, Commonwealth Hall) with licences in data/ATTRIBUTION.md, and experiment tables in report/results/.

The implementation does not call cv2.Stitcher, YOLO, or any pretrained recogniser.

2. App walkthrough

Six tabs: How it works, Build a panorama, Compare detectors, Robustness lab, Saved results, Project report.

3. Method (source excerpts)

Shared code is in src/. The app never duplicates the geometry: both the UI and scripts/run_experiments.py call match_pair / stitch_images.

Detect and describe. SIFT and ORB, capped at 2000 keypoints.

```
def create_detector(name, n_features=2000):
    key = name.strip().upper()
    if key == "SIFT":
        return cv2.SIFT_create(nfeatures=n_features)
    if key == "ORB":
        return cv2.ORB_create(nfeatures=n_features)
```

Match. Brute-force kNN (k=2) and Lowe's ratio test. SIFT uses L_2; ORB uses Hamming.

```
knn = matcher.knnMatch(descriptors_a, descriptors_b, k=2)
for best, second in knn:
    if best.distance < ratio * second.distance:
        good.append(best)
```

RANSAC homography. cv2.findHomography with a 5 px threshold. Pairwise maps are composed into the middle image's frame.

```
H, mask = cv2.findHomography(
    pts_src, pts_dst, method=cv2.RANSAC,
    ransacReprojThreshold=ransac_thresh)
```

Stitch. warpPerspective plus distance-transform feather blending. Metrics: inlier ratio, mean reprojection error, overlap SSIM.

4. Evidence

Primary scene: three Balme Library photographs (Wikimedia Commons, CC BY 4.0). Pair 2–3 is a near-duplicate facade (SIFT 1426 inliers, SSIM 0.976).

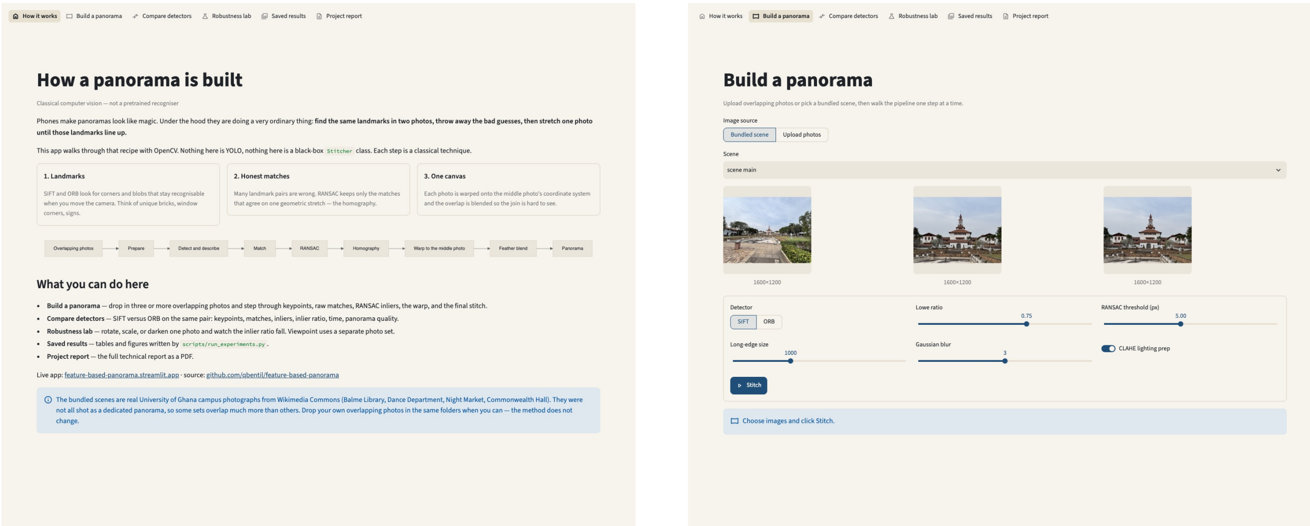

Figure 1. Left: How it works (pipeline explanation). Right: Build a panorama with bundled Balme Library views, SIFT, Lowe ratio 0.75, RANSAC 5 px.

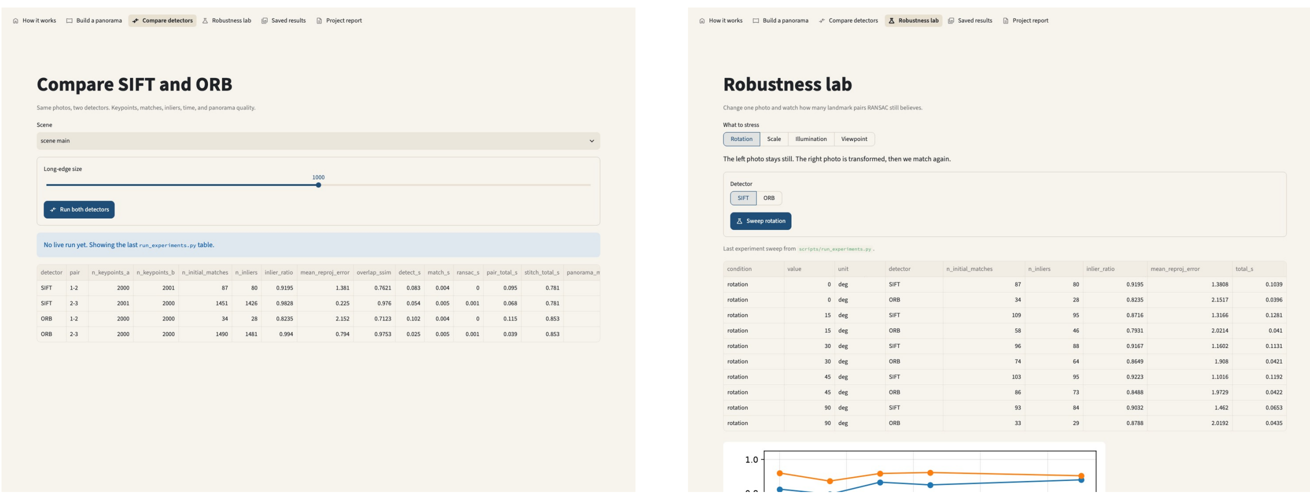

Figure 2. Left: Compare detectors (SIFT vs ORB table). Right: Robustness lab (rotation sweep).

Det.	Pair	Init.	Inliers	Ratio	SSIM
SIFT	1–2	87	80	0.920	0.762
SIFT	2–3	1451	1426	0.983	0.976
ORB	1–2	34	28	0.824	0.712
ORB	2–3	1490	1481	0.994	0.975

Table 1. Detector comparison on Balme Library consecutive pairs.

Pair 1–2 is courtyard-to-facade (80 inliers, SSIM 0.762, reprojection 1.38 px). Mean inlier ratio: SIFT 0.951, ORB 0.909. Full stitch about 0.8 s. Gain 0.35 illumination leaves SIFT with zero inliers. Dance Department viewpoint frames barely overlap.

5. How to reproduce

```
pip install -r requirements.txt
python scripts/run_experiments.py
streamlit run streamlit_app.py
```

Or open the live URL above. Bundled photos can be replaced with phone captures (30–50% overlap, left to right) in data/scenes/.

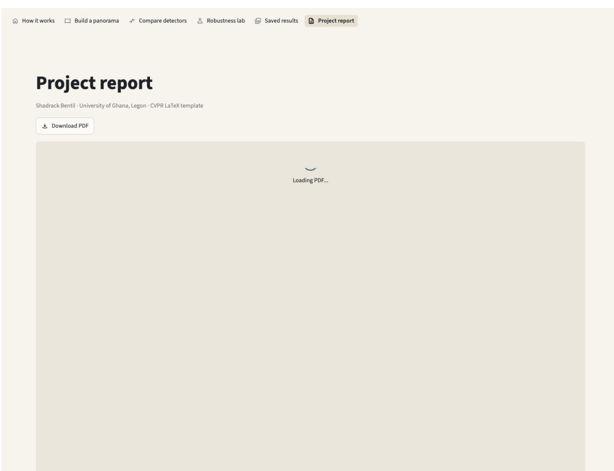

Figure 3. Project report tab: download and embed the full CVPR PDF.

Figure 4. Input views for the primary stitch (Balme Library).

Figure 5. Left: SIFT inliers after RANSAC. Right: stitched panorama.

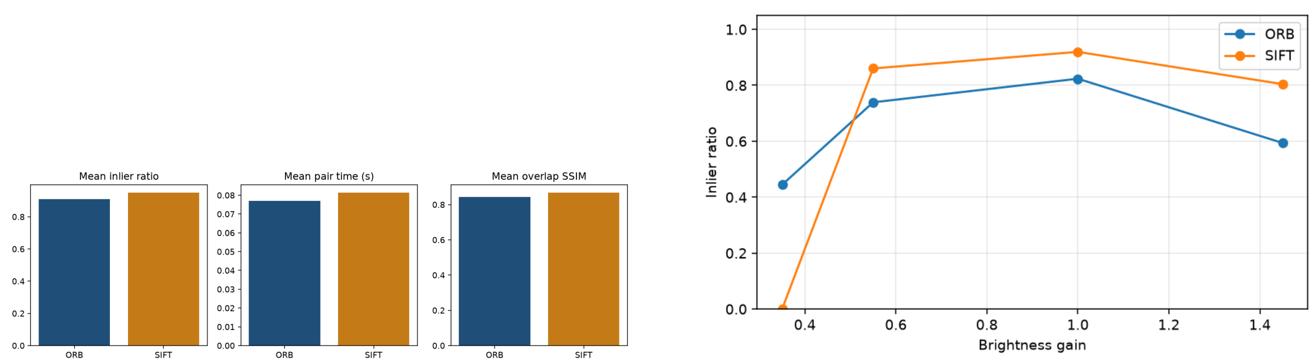

Figure 6. Left: SIFT vs ORB summary. Right: illumination sweep (inlier ratio vs brightness gain).